AMI Meter
Residential

E350

U1300 SERIES 1, SERIES 2, U3400

RELEASE NOTES FOR F/W VERSION R4.9

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Important Information

This document provides information relating to E350 U1300 Series 1, Series 2 and U3400 meter firmware (the "Product")

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1. About This Document

1.1. Purpose

The purpose of the Release document is to communicate major new features and fixes of R4.9 firmware release for E350 - U1300 S1, U1300 S2 and U3400 meters. It also documents known limitations.

1.2. Definitions, Acronyms, and Abbreviations

LC: Load Control
CLC/R: Controlled Load Contactor or Relay
QoS: Quality of Supply
LP: Load Profile
AMI: Advanced Metering Infrastructure
SCC: Supply Capacity Control
RSM: Relay State Machine
PP: Partial Programming
S1: Series 1
S2: Series 2

1.3. References

E350 U1300 Series User Manuals
E350 U3400 Series User Manuals

2. About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released Firmware	**This Released**	**Compatible Meter**	**Upgradability***	**Downgradability***
R4.8	**R4.9**	U1300 Series 1	3.2 → 4.9 ↓ 4.2 → 4.9 ↓ 4.3 → 4.9 ↓ 4.4 → 4.9 ↓ 4.5 → 4.9 ↓ 4.6 → 4.9 ↓ 4.8 → 4.9	4.9 → 4.8 ↓ 4.9 → 4.6 ↓ 4.9 → 4.5 ↓ 4.9 → 4.4 ↓ 4.9 → 4.3 ↓ 4.9 → 4.2
R4.8	**R4.9**	U1300 Series 2	4.4 → 4.9 ↓ 4.5 → 4.9 ↓ 4.6 → 4.9 ↓ 4.8 → 4.9	4.9 → 4.8 ↓ 4.9 → 4.6 ↓ 4.9 → 4.5 ↓ 4.9 → 4.4
R4.6	**R4.9**	U3400	4.4 → 4.9 ↓ 4.5 → 4.9 ↓ 4.6 → 4.9	4.9 → 4.6 ↓ 4.9 → 4.5 ↓ 4.9 → 4.4

Important Note: *customer will be required to refer to the release notes for the particular firmware version to understand the limitations of the upgrade/downgrade. Please contact your sales representative for more information.

Broadly, this R4.9 firmware release provides the following information features of U1300 S1, S2 and U3400 meters:

(i) Support for Priority Override Commands
(ii) Demand Load Limiting Control (DLLC) Switching
(iii) Enhanced Supply Capacity Control

This firmware has been tested on the following platforms:

- E350 U1300 meters (this includes Series 1 and Series 2)
- E350 U3400 meters
- EMPWin+ 2.12

3. New Features, Important Notes and Fixes

3.1. **New Feature:** Support for Priority Override Commands

3.1.1. General Details – Priority Override Commands

Priority Override Commands are commands that are sent from the remote head end system (or NMS) to the meter. The purpose of these commands is to override the state of the CLC/R of the meter. It shall be noted that Priority Override Commands can only override RSM modules that have a lower priority than the Priority Override Command.

The head end system (or NSM) would need to implement the Priority Override Commands so that it can be used with L+G E350 U1300 and U3400 meters. EMPWin+ can also be used to send these commands from its toolbox menu when connected to the meter (remotely or locally).

In R4.9 firmware, the meter supports the following new Priority Override Commands for CLC/R Operation:

i. Turn ON Immediately
ii. Turn OFF Immediately
iii. Delayed Turn ON (using Switch Time Randomisation provided in Command)
iv. Delayed Turn OFF (using Switch Time Randomisation provided in Command)
v. Start load cycling

For termination of above mentioned commands, meters with R4.9 firmware now also support Terminate Priority Override Command as explained in a later section of this document.

Priority Override Commands are available in addition to existing Legacy Commands (Turn ON and Turn OFF). The Legacy Commands are kept in R4.9 firmware to maintain backward combability.

For the Priority Override Commands, after the expiry of Override Duration, the RSM releases control to any other module based on the Priority table as shown in Section 3.4 of this document.

3.1.1.1 Delayed Turn ON and Delayed Turn OFF Commands

The Delayed Turn ON and Delayed Turn OFF Commands use a Random Delay, in order to avoid synchronised switching of CLC/R on multiple meters that receive the Command. Hence E350 meter with R4.9 firmware accepts Commands with a minimum and maximum time delay of 0 to 720 minutes (12 hours) in steps of 1 minute.

3.1.1.2 Start Load Cycling commands

When the Start load control cycling program starts;

a. If the initial position of the CLC/R is in the OFF position, then after the Random Delay the CLC/R will switch to the ON position and will commence cycling (OFF and ON) in accordance with the cycle time defined in the meter.
b. If the initial position of the CLC/R is in the ON position then after the Random Delay the CLC/R will switch to the OFF position and will commence cycling (ON and OFF) in accordance with the cycle times defined in the meter.

3.1.1.3 Terminate Priority Override Command

The Terminate Priority Override Command supports settings for:

a. The CLC/R that is being controlled.
b. Either to end after a Random Delay (minimum delay is fixed to 0 and maximum delay can be 120 minutes) or to end immediately (making maximum delay as 0 min). This Random delay is used, only if Priority Override Command received by the meter has already taken control of the RSM.

Operation of the terminate Priority Override Command:

a. If the Terminate Priority Override Command is received before the specified Start Date & Time: the Priority Override is cancelled and the CLC/R does not commence Priority Override.
b. If the Terminate Priority Override Command is received during the Priority Override: the CLC/R releases control to the RSM, either after a Random Delay or immediately as specified in the terminate Priority Override Command.
c. If the Terminate Priority Override Command is received when no Priority Override is programmed (including Legacy Priority Override): the CLC/R continues to use the current mode.

3.2. **New Feature:** Demand Limit Load Control Switching

R4.9 firmware supports Demand Limit Load Control (DLLC) switching of the CLC/R as per following details:

The meter allows configuration of the following:

a. DLLC switching to be enabled or disabled.
b. The start time for the DLLC switching to be set in hours and minutes.
c. The duration for the DLLC switching to be set in minutes from 5 to 1440 minutes (24 hours) in increments of 5 minutes.
d. The demand limit supply capacity limit to be set from 0.1 kWh to 99 kWh in 0.1 kWh increments.
e. The measurement period to be set from 1 to 60 minutes in 1 minute intervals.
f. The position of the CLC/R during DLLC switching as either turned on or turned off.
g. The meter to automatically initiate DLLC switching while the Emergency Supply Capacity Limit is initiated as enabled or disabled.
h. The DLLC cannot be selected individually for each season. To disable DLLC functionality for a particular season, the kWh limit for that particular season can be set to maximum (99 kWh).

For DLLC, the measurement examples are shown in Annex 1 attached at the end of the release notes.

IMPORTANT NOTE: EMPWin+ 2.12 version will have support of DLLC feature of R4.9. It shall be noted that older firmware (R4.8 and older) also has DLLC features (except the linkage between Emergency Supply Capacity Control and DLLC), however it wasn't officially released, and hence it is not recommended to use DLLC features on meters with R4.8, as the functionality will not be guaranteed.

3.2.1. Operation of Demand Limit Load Control switching:

If DLLC switching is enabled and the meter time indicates that DLLC switching should be initiated then:

a. Starting at the programmed start time the meter will measure the total imported active energy in the measurement period.
   i. If the measurement exceeds the "Demand Limit Supply Capacity" limit, the meter will switch the CLC/R to the programmed position for the remainder of the measurement period. At the end of the measurement period control of the CLC/R returns to the highest operational load control mode. An event will be written to the meter's quality of supply event log indicating that the CLC/R has switched due to DLLC.
   ii. If the measurement does not exceed the programmed "Demand Limit Supply Capacity" limit, then the meter will not switch the CLC/R and will repeat the energy measurement for each subsequent measurement period.
   iii. At the conclusion of the programmed duration, control of the CLC/R returns to the highest operational load control mode.

There is a configurable setting in the DLLC page of Empwin+ 2.12, to initiate (or not initiate) the Demand Limit Load Control, when ESCC is enabled.

3.3. **New Feature:** Enhanced Supply Capacity Control

R4.9 firmware's supply capacity control feature now supports five supply capacity control limits:

- Time of Day (TOD) Import Supply Capacity Limit
- Time of Day (TOD) Export Supply Capacity Limit
- Distribution Network Import Supply Capacity Limit
- Distribution Network Export Supply Capacity Limit
- Emergency Supply Capacity Limit

3.3.1. Time of Day Import Supply Capacity Limit

Time of Day Import Supply Capacity Limit is the same as the existing Normal SCC implementation of R4.8, with enhancements as per below:

Time of day can only be on half hour boundary as per meter RTC.

The start time for the Time of Day Import Supply Capacity Limit to be set in hours and minutes.

The duration for the Time of Day Import Supply Capacity Limit to be set in minutes from 5 to 1440 minutes (24 hours) in increments of 5 minutes.

Other features of SCC will remain same as existing normal Supply Capacity Control implementation of R4.8, including enabling and disabling of Time of Day Import Supply Capacity Limiting, Re-Close Delay, Import Demand duration, Import Demand limit. Notes that this is based on Clause 3.9.1 of DPI (department of Primary Industries – Minimum AMI Functionality Specification (Victoria) – September 2008, Release 1.1.

3.3.2. Time of Day Export Supply Capacity Limit

This implementation is same as "Time of Day Import Supply Capacity Limit" described above, except the direction of flow of energy, i.e. Export (instead of Import).

3.3.3. Distribution Network Import Supply Capacity Limit

Distribution Network Import Supply Capacity Limit is the same as the existing Normal Supply Capacity Control implementation of R4.8

In Empwin+ 2.12 the settings still refer to Normal Supply Capacity Limit.

3.3.4. Distribution Network Export Supply Capacity Limit

This implementation is same as “Distribution Network Import Supply Capacity Limit” described above, except the direction of flow of energy, i.e. Export (instead of Import).

3.3.5. Emergency Supply Capacity Limit

Emergency Supply Capacity Limit is the same as existing Emergency Supply Capacity Control implementation of R4.8.

As stated in Section 3.2.1, there is a configurable setting in the DLLC page of Empwin+ 2.12, to initiate (or not initiate) the Demand Limit Load Control, when ESCC is enabled. This is a new feature in R4.9.

3.3.6. Local and Remote enabling and disabling of Supply Capacity Limit.

If a meter has R4.9 firmware, with TOD Supply Capacity controls enabled (at that point in time), and then if this meter is downgraded to R4.8 or below, then the Supply Capacity Control will not work. In this case meter will have to be reprogrammed (or remote Command will be required) for SCC features to work.

However, if a meter has R4.9 firmware, with TOD Supply Capacity controls not enabled (at that point in time), and then if this meter is downgraded to R4.8 or below, then the Supply capacity control will work as per the capability of the firmware.

3.4. CLC/R State Priorities

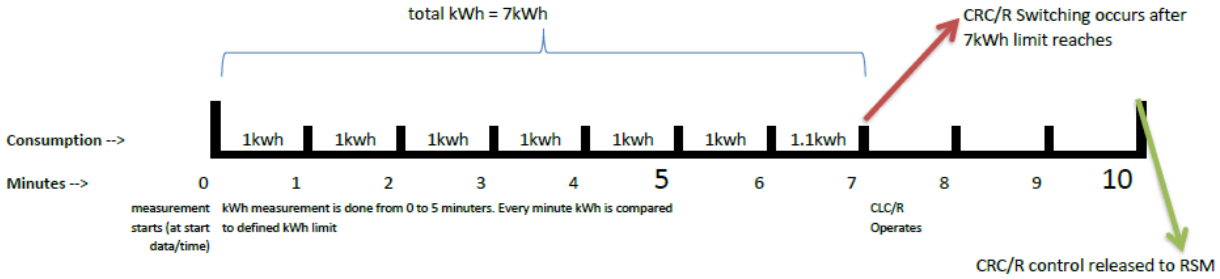
The updated CLC/R State Priorities based on features defined in Section 3.1 and 3.2 of this document are:

- Priority 1 Disconnect on Loss of Supply
- Priority 2 Under Frequency Load Switching
- Priority 3 Under Voltage (not released)
- Priority 4 Boost with Boost Primacy enabled
- **Priority 5 Priority Override**
- **Priority 6 Demand Limit**
- Priority 7 Boost with Boost Primacy disabled
- Priority 8 Ripple Control
- Priority 9 Power Restore
- Priority 10 Time Table

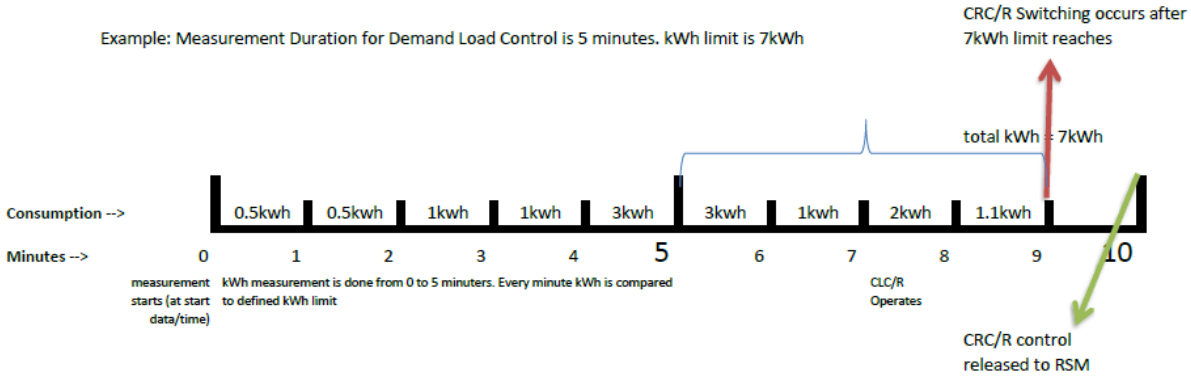
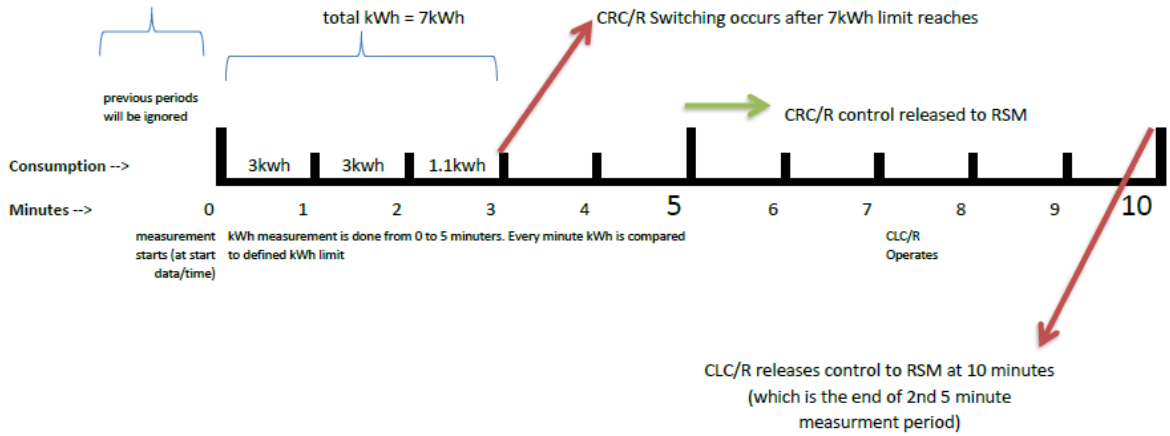
ANNEX 1

Demand Load Limit Control : Fixed Windows Measurement Examples

Example: Measurement Duration for Demand Load Control is 10 minutes. kWh limit is 7kWh



Example: Measurement Duration for Demand Load Control is 5 minutes. kWh limit is 7kWh



Example: Measurement Duration for Demand Load Control is 5 minutes. kWh limit is 7kWh